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Power Outage Forecasting and Decision-Making

S26 95-828 Machine Learning for Problem Solving

Motivation: Pittsburgh outage, May 2025

Not able to fix the problem completely

April 30
140K outage
in Pittsburgh

12 hours later
84K outage
400 electricians
on the ground

24 hours later
Ongoing outage

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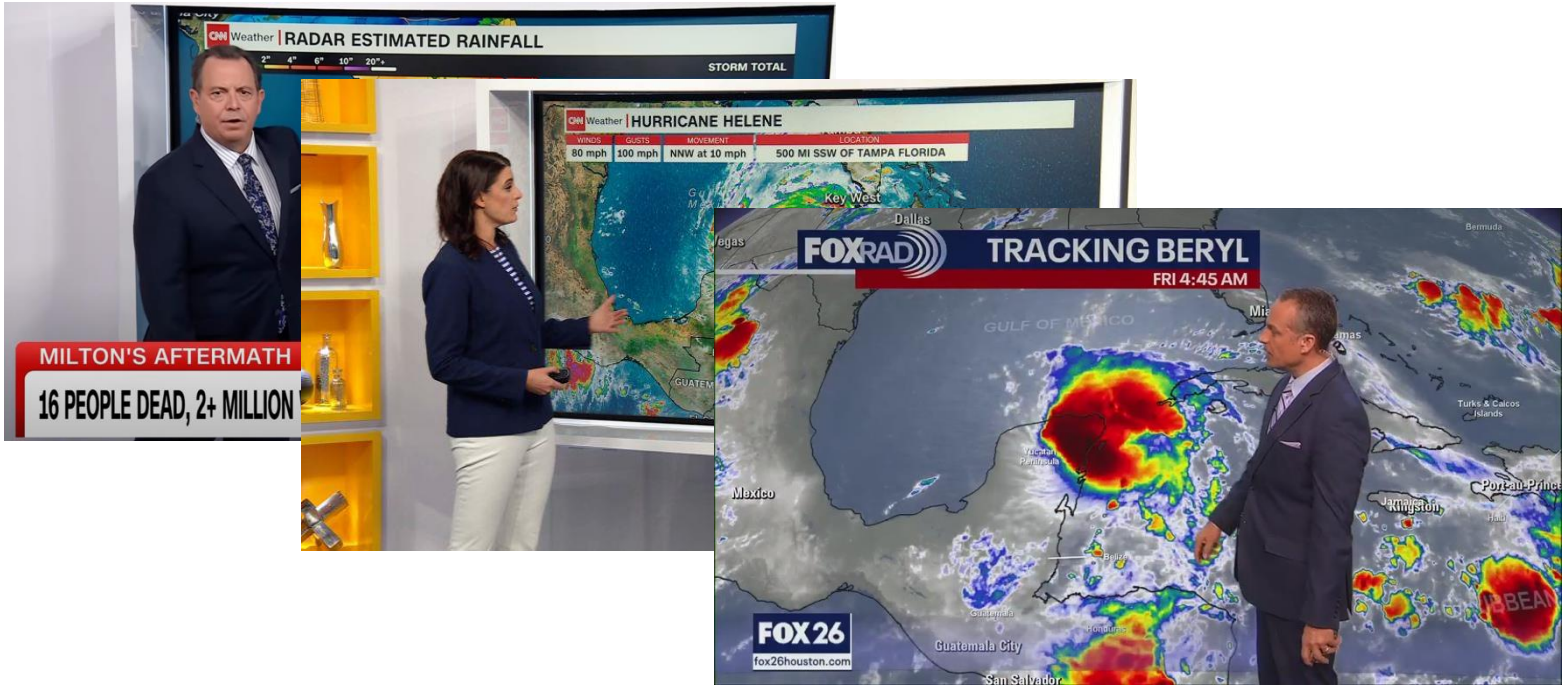
Motivation: Evidence across the nation



[1] Florida Power & Light. “FPL Continues to Strengthen Energy Grid Following Record Hurricane Season.” Newsroom release, Nov. 27, 2024.
[2] Utility Dive. “Hurricane Milton Cuts Power to More Than 3 Million Across Florida.” Oct. 10, 2024.

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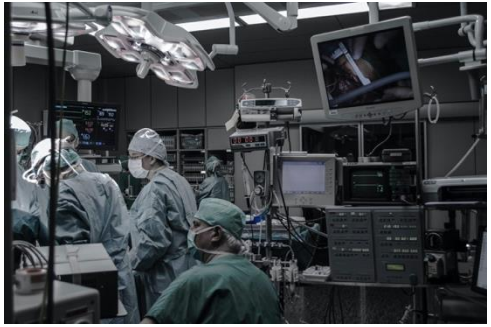
Motivation: Significance of Power Grid Resilience

Resilience - *the capability of withstanding, adapting to, and recovering from a large-scale disruption* - has become a top priority for the power sector.



Public Safety

Economic stability

National Security

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Motivation: Proactive Action

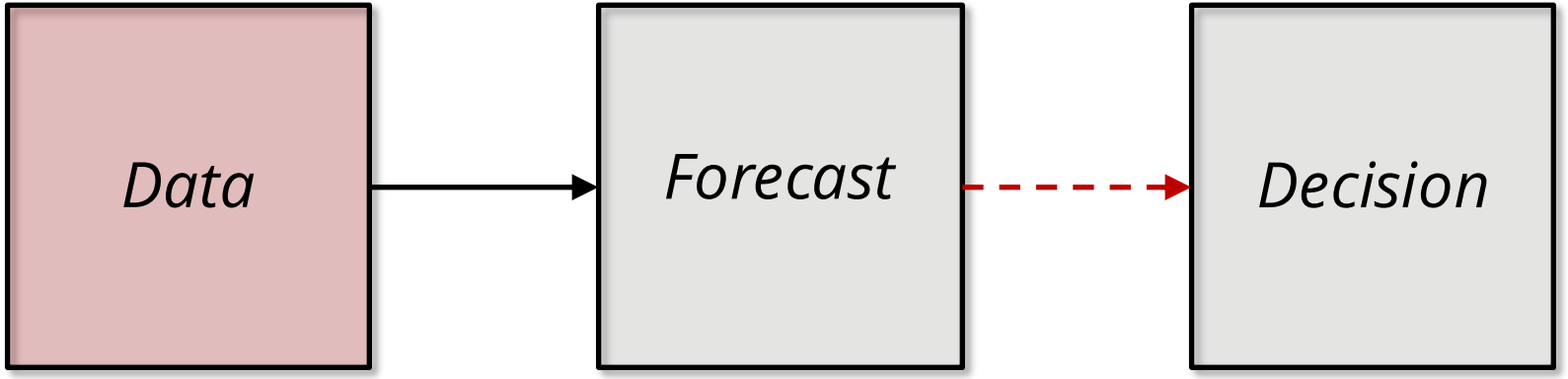
- Proactive actions (ex ante): decision made before disasters to reduce the likelihood or severity of future outages.

Power line undergrounding

Hardening the poles

Vegetation management

Problem Description



- *Outage data*
- *Weather data*

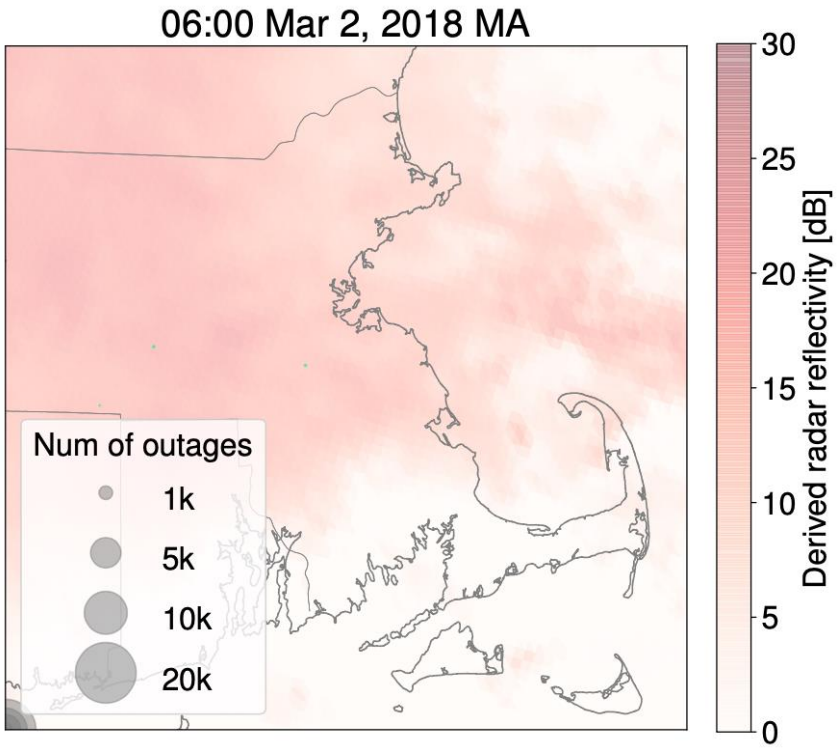
- *Prediction Model*

- *Policy Recommendation*

Dataset

- The dataset contains hourly county-level outage counts and weather variables for all **83** counties in Michigan (FIPS codes 26001–26165).
- The training period is from 2023-04-01 00:00 to 2023-06-30 00:00. The testing period is the following **24** and **48 hours**.
- The dataset contains **109** weather variables, covering temperature, humidity, wind, precipitation, severe weather, and other atmospheric and land surface conditions.
- To mirror real deployment environment, weather variables during testing periods will be provided nor should be used.

Dataset: Visualization (demo)



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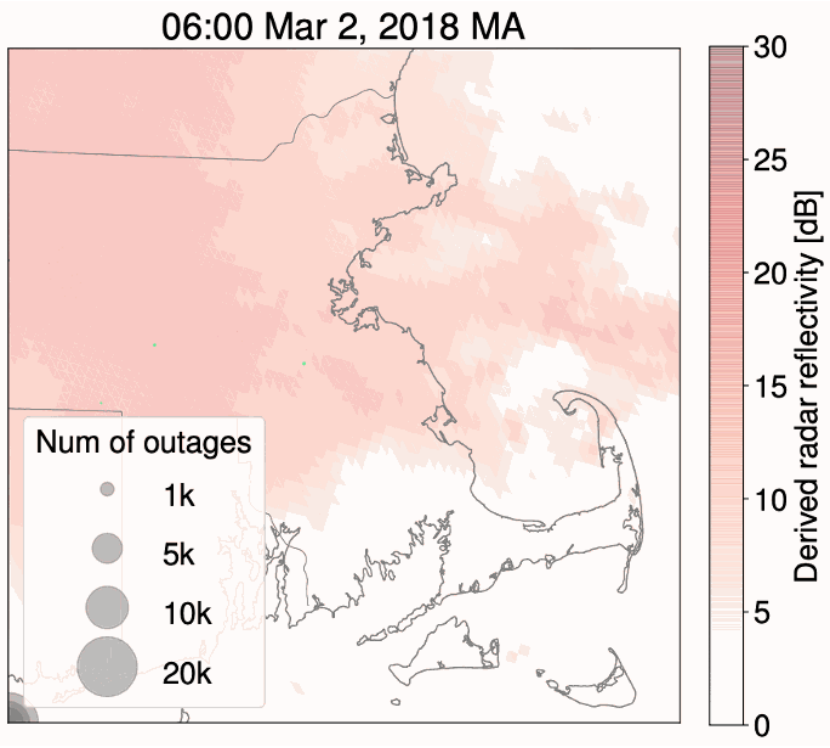
Dataset: Visualization (demo)



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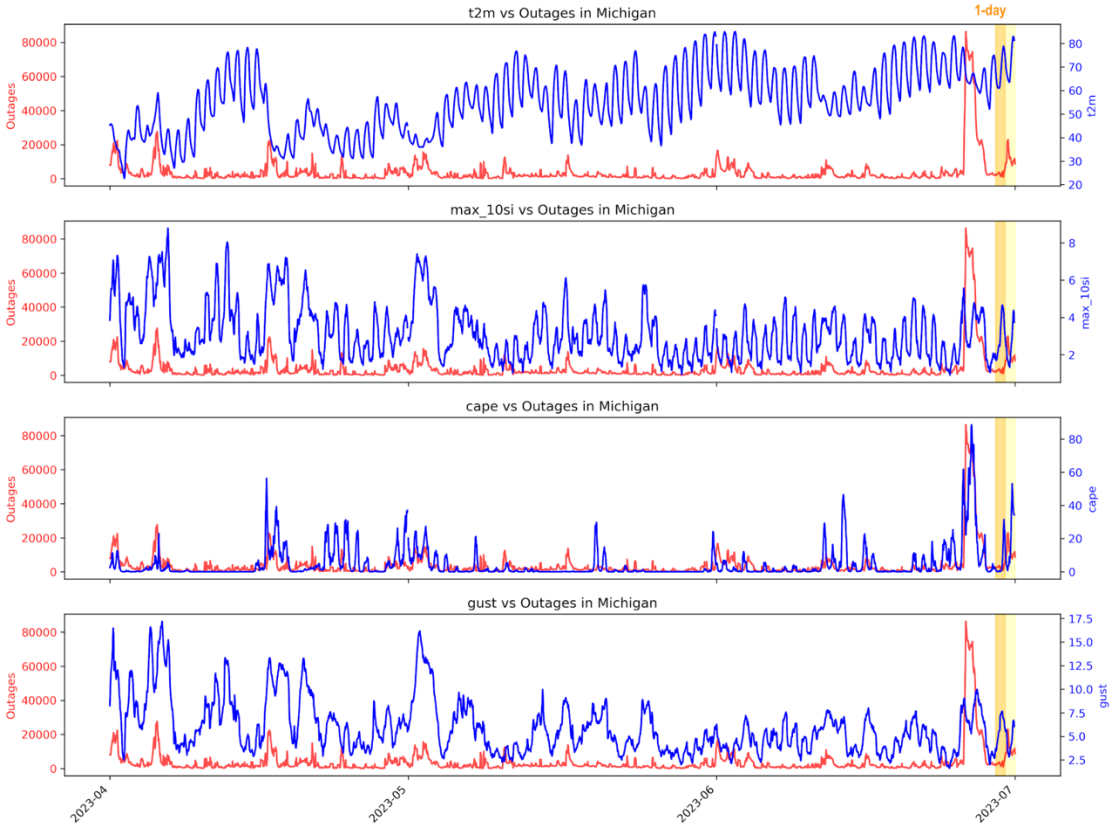
Dataset: Weather Data and Outages



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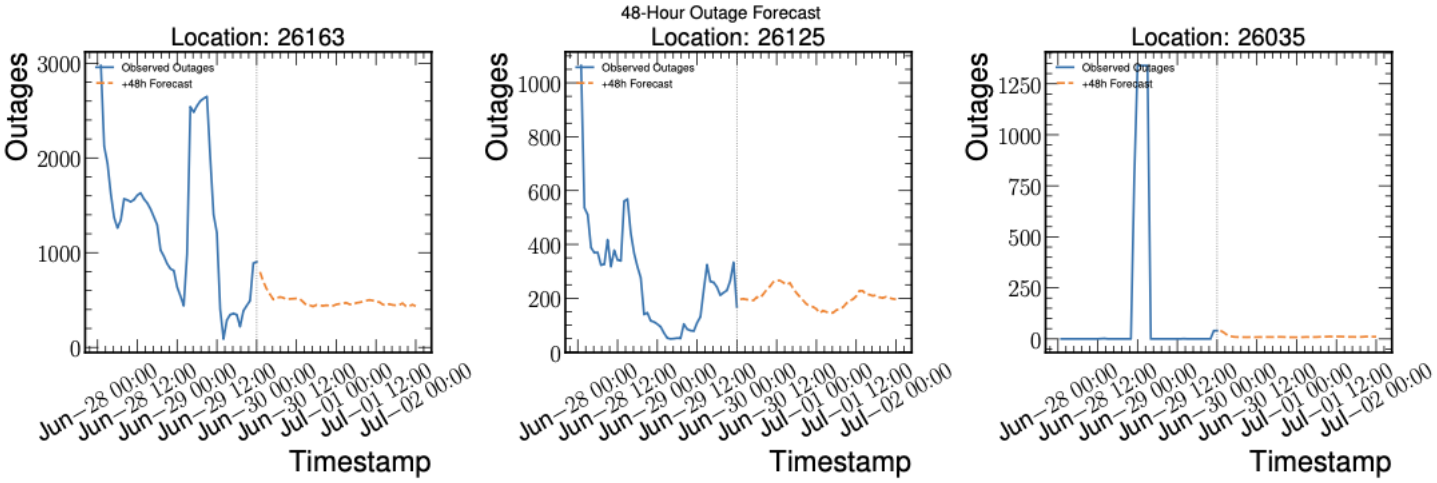
Dataset: Outage Prediction



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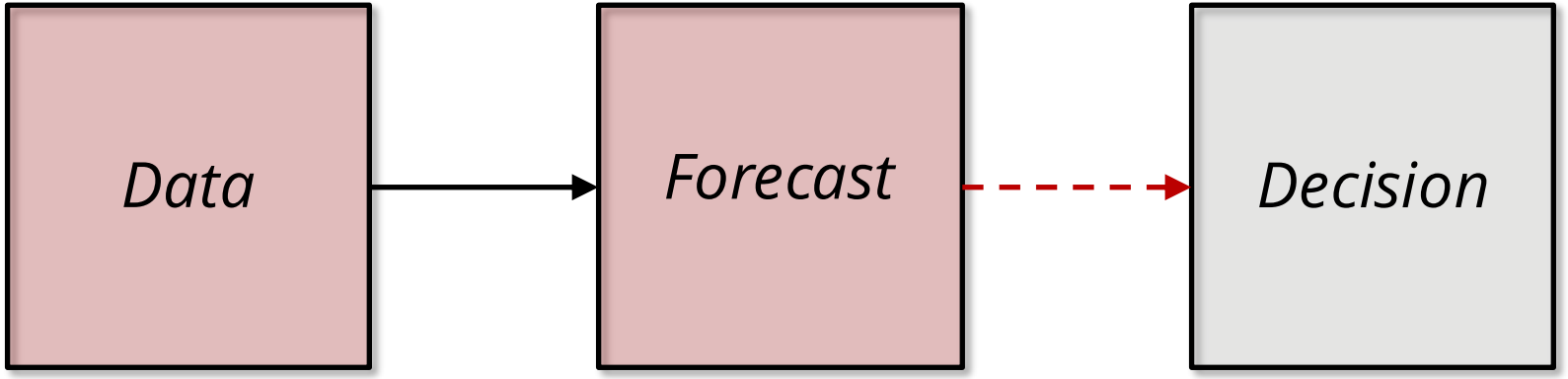
Problem Description



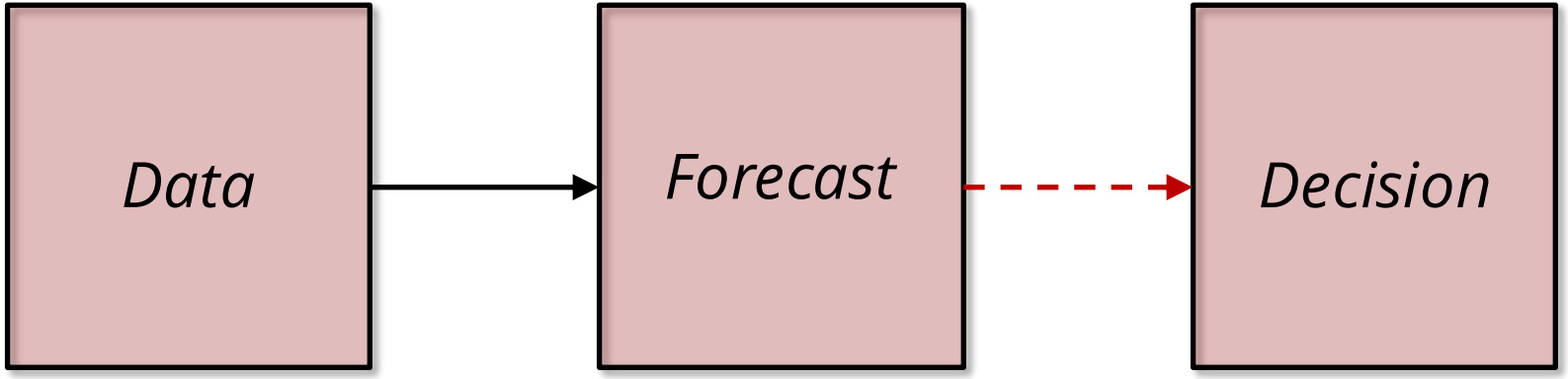
- *Outage data*
- *Prediction Model*
- *Policy Recommendation*

Part I: Short-term Outage Prediction

- **Input** of your prediction model:
  - Historical weather features
  - Could use techniques like PCA to reduce dimensionality

- **Output** of your prediction model:
  - Hourly prediction over 24 hours and 48 hours following training period at county level

Problem Description



- *Outage data*
- *Prediction Model*
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Part II: Policy Recommendations based on Prediction

Context:

A utility is trying to decide how to allocate its five mobile generators to prepare for forecasted outage. Each generator can provide power for 1000 households. And once a generator is assigned to a county, it cannot be moved.

Output:

You should recommend five counties to utility where to allocate those five generators.

Mobile generators

Evaluation and Grading Rubric

Criterion	Weight
Prediction Performance	**30%**
24-hour Prediction RMSE	15%
48-hour Prediction RMSE	15%
Quality of the Report	**70%**
Writing quality, clarity, and organization	10%
Modeling methodology and justification	15%
Evaluation and discussion of prediction results	15%
Policy recommendation and justification based on prediction	30%

Evaluation and Grading Rubric

- **Prediction Evaluation (30%)**

Your predictions are generated by a learned forecasting model. With prediction and true data, the RMSE is defined as:

$$\mathrm{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(y_t - \hat{y}_t)^2},$$

where T = 24 or 48. The final performance metric is the average RMSE across all counties.

Evaluation and Grading Rubric

- **Policy Recommendation Evaluation (30%)**

Your recommendation should be a vector of five counties where you recommend utility to allocate its five generators.

It will be evaluated based on

- Justification your recommendation based on data analysis and prediction
- The actual quality of your recommendation (we will evaluate it based on real data)

Evaluation and Grading Rubric

- Report Writing

1. **Introduction**: Clearly describe the decision context and the prediction task. Explain why the problem matters from the perspective of a decision maker. Briefly summarize the dataset, key features, and prediction targets.
2. **Data Analysis**: Describe exploratory data analysis, including data cleaning, feature transformations, and key patterns or challenges identified in the data.
3. **Methods**: Describe the models implemented and the rationale for choosing them. Discuss baseline approaches that were implemented and compared to.
4. **Results**: Present experimental results using appropriate visualizations and tables. Clearly interpret results and relate them to expectations and modeling choices.
5. **Analysis and Policy Recommendation**: Analyze results from a decision-making perspective. Discuss limitations, uncertainty, and assumptions. Clearly justify policy recommendations using evidence from the predictive analysis.
6. **References and Citations**: Include properly formatted references for all external sources and tools used.

Deliverables and Submission Format

By April 26th at 11:59PM, you should submit:

1. A single project paper (PDF), maximum 8 pages excluding references and appendices via Canvas;
2. A single Jupyter notebook containing all code and results via Canvas;
3. Two prediction files and a decision file submitted via a Google Form

Deliverables and Submission Format

Submission Format:

Prediction file (.csv):

Table 1: Expected format for prediction file submissions.

timestamp	location	pred
2023-06-30 01:00	26001	0
2023-06-30 02:00	26001	0
⋮	⋮	⋮

Decision file (.txt):

```
[26001, 26001, 26001, 26001, 26001]
```

Timeline

- **Jan 22rd: Project Release**
  Problem statement and dataset are released. Begin exploring the dataset, examining basic statistics and visualizations, understanding the prediction task, and identifying a preliminary policy question. Start early!

- **Feb 20th at 11:59 PM: Team Information Deadline**
  Form teams of up to four students. Teams of two or three are also acceptable. Team details will be collected with a Google Form.

- **April 26th at 11:59 PM: Final Submission Deadline**
  Submit Jupyter notebook and project report (PDF) via Canvas, and prediction files and recommendation via Google Form.